Cutoff-Date Sensitivity — Modeled

Savings = log-linear regression baseline (LITERATURE_ANNUAL_CUTS, median total ET, WY type) × literature cut-timing scaling factor.

Cutoff date Jul 1 → Oct 1 | italic = all values from calibrated model

Late cuts — count

Cap 0 — remove all late cuts

Cap 1 — keep 1 late cut

(a) Critical

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	3	3	2	2	1	0	0
Riverside	3	3	2	2	1	0	0
Kern	3	2	2	1	1	0	0
Kings	2	2	2	1	1	0	0
Tulare	2	2	2	1	1	0	0
Fresno	2	2	2	1	1	0	0
Madera	2	2	1	1	1	0	0
Merced	2	2	1	1	1	0	0
Stanislaus	2	2	1	1	1	0	0
S. Joaquin	2	2	1	1	1	0	0

(b) Dry

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	3	3	2	2	1	0	0
Riverside	3	3	2	2	1	0	0
Kern	3	2	2	1	1	0	0
Kings	2	2	2	1	1	0	0
Tulare	2	2	2	1	1	0	0
Fresno	2	2	2	1	1	0	0
Madera	2	2	1	1	1	0	0
Merced	2	2	1	1	1	0	0
Stanislaus	2	2	1	1	1	0	0
S. Joaquin	2	2	1	1	1	0	0

(c) Above Normal

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	3	3	2	2	1	0	0
Riverside	3	3	2	2	1	0	0
Kern	3	2	2	1	1	0	0
Kings	2	2	2	1	1	0	0
Tulare	2	2	2	1	1	0	0
Fresno	2	2	2	1	1	0	0
Madera	2	2	1	1	1	0	0
Merced	2	2	1	1	1	0	0
Stanislaus	2	2	1	1	1	0	0
S. Joaquin	2	2	1	1	1	0	0

(d) Wet

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	3	3	2	2	1	0	0
Riverside	3	3	2	2	1	0	0
Kern	3	2	2	1	1	0	0
Kings	2	2	2	1	1	0	0
Tulare	2	2	2	1	1	0	0
Fresno	2	2	2	1	1	0	0
Madera	2	2	1	1	1	0	0
Merced	2	2	1	1	1	0	0
Stanislaus	2	2	1	1	1	0	0
S. Joaquin	2	2	1	1	1	0	0

Mean late cuts (n): 0.0 – 0.5 – 1.0 – 1.5 – 2.0 – 2.5 – 3.0

(e)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.66	1.66	1.11	1.11	0.55	0.00	0.00
Riverside	1.72	1.72	1.14	1.14	0.57	0.00	0.00
Kern	0.99	0.66	0.66	0.33	0.33	0.00	0.00
Kings	1.04	1.04	1.04	0.52	0.52	0.00	0.00
Tulare	1.00	1.00	1.00	0.50	0.50	0.00	0.00
Fresno	1.04	1.04	1.04	0.52	0.52	0.00	0.00
Madera	1.05	1.05	0.53	0.53	0.53	0.00	0.00
Merced	1.06	1.06	0.53	0.53	0.53	0.00	0.00
Stanislaus	0.95	0.95	0.47	0.47	0.47	0.00	0.00
S. Joaquin	0.89	0.89	0.44	0.44	0.44	0.00	0.00

(f)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.64	1.64	1.09	1.09	0.55	0.00	0.00
Riverside	1.69	1.69	1.13	1.13	0.56	0.00	0.00
Kern	0.98	0.65	0.65	0.33	0.33	0.00	0.00
Kings	1.03	1.03	1.03	0.51	0.51	0.00	0.00
Tulare	0.98	0.98	0.98	0.49	0.49	0.00	0.00
Fresno	1.03	1.03	1.03	0.51	0.51	0.00	0.00
Madera	1.04	1.04	0.52	0.52	0.52	0.00	0.00
Merced	1.05	1.05	0.52	0.52	0.52	0.00	0.00
Stanislaus	0.94	0.94	0.47	0.47	0.47	0.00	0.00
S. Joaquin	0.88	0.88	0.44	0.44	0.44	0.00	0.00

(g)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.57	1.57	1.05	1.05	0.52	0.00	0.00
Riverside	1.62	1.62	1.08	1.08	0.54	0.00	0.00
Kern	0.94	0.62	0.62	0.31	0.31	0.00	0.00
Kings	0.98	0.98	0.98	0.49	0.49	0.00	0.00
Tulare	0.94	0.94	0.94	0.47	0.47	0.00	0.00
Fresno	0.99	0.99	0.99	0.49	0.49	0.00	0.00
Madera	0.99	0.99	0.50	0.50	0.50	0.00	0.00
Merced	1.00	1.00	0.50	0.50	0.50	0.00	0.00
Stanislaus	0.90	0.90	0.45	0.45	0.45	0.00	0.00
S. Joaquin	0.84	0.84	0.42	0.42	0.42	0.00	0.00

(h)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.73	1.73	1.15	1.15	0.58	0.00	0.00
Riverside	1.79	1.79	1.19	1.19	0.60	0.00	0.00
Kern	1.03	0.69	0.69	0.34	0.34	0.00	0.00
Kings	1.09	1.09	1.09	0.54	0.54	0.00	0.00
Tulare	1.04	1.04	1.04	0.52	0.52	0.00	0.00
Fresno	1.09	1.09	1.09	0.54	0.54	0.00	0.00
Madera	1.09	1.09	0.55	0.55	0.55	0.00	0.00
Merced	1.11	1.11	0.55	0.55	0.55	0.00	0.00
Stanislaus	0.99	0.99	0.49	0.49	0.49	0.00	0.00
S. Joaquin	0.93	0.93	0.46	0.46	0.46	0.00	0.00

Cap 0 savings (ac-ft/acre): 0.0 – 0.2 – 0.4 – 0.6 – 0.8 – 1.0 – 1.2 – 1.4 – 1.6

(i)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.12	1.12	0.56	0.56	0.00	0.00	0.00
Riverside	1.17	1.17	0.58	0.58	0.00	0.00	0.00
Kern	0.55	0.28	0.28	0.00	0.00	0.00	0.00
Kings	0.53	0.53	0.53	0.00	0.00	0.00	0.00
Tulare	0.51	0.51	0.51	0.00	0.00	0.00	0.00
Fresno	0.53	0.53	0.53	0.00	0.00	0.00	0.00
Madera	0.48	0.48	0.00	0.00	0.00	0.00	0.00
Merced	0.49	0.49	0.00	0.00	0.00	0.00	0.00
Stanislaus	0.43	0.43	0.00	0.00	0.00	0.00	0.00
S. Joaquin	0.40	0.40	0.00	0.00	0.00	0.00	0.00

(j)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.16	1.16	0.58	0.58	0.00	0.00	0.00
Riverside	1.21	1.21	0.60	0.60	0.00	0.00	0.00
Kern	0.57	0.29	0.29	0.00	0.00	0.00	0.00
Kings	0.55	0.55	0.55	0.00	0.00	0.00	0.00
Tulare	0.52	0.52	0.52	0.00	0.00	0.00	0.00
Fresno	0.55	0.55	0.55	0.00	0.00	0.00	0.00
Madera	0.50	0.50	0.00	0.00	0.00	0.00	0.00
Merced	0.50	0.50	0.00	0.00	0.00	0.00	0.00
Stanislaus	0.45	0.45	0.00	0.00	0.00	0.00	0.00
S. Joaquin	0.42	0.42	0.00	0.00	0.00	0.00	0.00

(k)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	0.75	0.75	0.38	0.38	0.00	0.00	0.00
Riverside	0.78	0.78	0.39	0.39	0.00	0.00	0.00
Kern	0.37	0.18	0.18	0.00	0.00	0.00	0.00
Kings	0.35	0.35	0.35	0.00	0.00	0.00	0.00
Tulare	0.34	0.34	0.34	0.00	0.00	0.00	0.00
Fresno	0.35	0.35	0.35	0.00	0.00	0.00	0.00
Madera	0.32	0.32	0.00	0.00	0.00	0.00	0.00
Merced	0.32	0.32	0.00	0.00	0.00	0.00	0.00
Stanislaus	0.29	0.29	0.00	0.00	0.00	0.00	0.00
S. Joaquin	0.27	0.27	0.00	0.00	0.00	0.00	0.00

(l)

County (N → S)	Jul 1	Jul 16	Aug 1	Aug 16	Sep 1	Sep 16	Oct 1
Imperial	1.18	1.18	0.59	0.59	0.00	0.00	0.00
Riverside	1.22	1.22	0.61	0.61	0.00	0.00	0.00
Kern	0.58	0.29	0.29	0.00	0.00	0.00	0.00
Kings	0.55	0.55	0.55	0.00	0.00	0.00	0.00
Tulare	0.53	0.53	0.53	0.00	0.00	0.00	0.00
Fresno	0.56	0.56	0.56	0.00	0.00	0.00	0.00
Madera	0.50	0.50	0.00	0.00	0.00	0.00	0.00
Merced	0.51	0.51	0.00	0.00	0.00	0.00	0.00
Stanislaus	0.45	0.45	0.00	0.00	0.00	0.00	0.00
S. Joaquin	0.42	0.42	0.00	0.00	0.00	0.00	0.00

Cap 1 savings (ac-ft/acre): 0.0 – 0.2 – 0.4 – 0.6 – 0.8 – 1.0 – 1.2

Cutoff date

Model: log(saved + 1) = β₀ + β_nc·log(nc) + β_ET·log(total ET) + γ_WY | Generalization gap = CV-RMSE / in-sample-RMSE

5-fold CV — Cap 0: N=10,693, MAE=132mm, RMSE=171mm, NRMSE=0.44, gap=1.000; Cap 1: N=10,693, MAE=91mm, RMSE=116mm, NRMSE=0.59, gap=1.001

Abbreviations: nc = number of cuttings; ET = evapotranspiration; WY = water year; ac-ft/acre = acre-feet per acre; obs = observation(s); MAE = mean absolute error; RMSE = root-mean-square error; NRMSE = RMSE / mean(observed savings); CV = cross-validation.